Inverse functions

1. Let f be a one-to-one differentiable function such that $f(3) = 7$, $f(7) = 8$, $f'(3) = 2$ and $f'(7) = 3$. The function g is differentiable and $g(x) = f^{-1}(x)$ for all x.

   $g'(7)$ is equal to

   A. $\frac{1}{2}$ B. 2 C. $\frac{1}{6}$ D. $\frac{1}{8}$ E. $\frac{1}{3}$

Inverse trig

Derivatives of inverse circular functions:

$$\frac{d}{dx}(\sin^{-1}(x)) = \frac{1}{\sqrt{1-x^2}}$$
$$\frac{d}{dx}(\cos^{-1}(x)) = \frac{-1}{\sqrt{1-x^2}}$$
$$\frac{d}{dx}(\tan^{-1}(x)) = \frac{1}{1+x^2}$$

1. Find the derivative of $8 \arcsin\left(\frac{x+1}{4}\right) + \frac{(x+1)\sqrt{-x^2-2x+15}}{2}$.

2. a. Find the derivative of $\sin^{-1}\left(\frac{2}{x-1}\right)$.

b. Find the maximal domain and range of $y = \sin^{-1}\left(\dfrac{2}{x-1}\right)$.

c. Sketch the graph of $y = \sin^{-1}\left(\dfrac{2}{x-1}\right)$ over its maximal domain. Label all endpoints and axes intercepts with their coordinates and all asymptotes with their equation.

Implicit differentiation

1. Find the equation of the normal to the curve defined by the relation $x^3 + y^3 - 2xy = 0$ at the point $(1, 1)$.

2. Consider the function $f : [-1, +\infty) \to R$, $f(x) = xe^x$. Find the gradient of the normal to the **inverse function** f^{-1} at the point where $x = e$.

Second derivatives and inflection points

- when is there a POI? Concavity changes (i.e f'' changes sign) in other words, f' has a maxima/minima.

1. For the curve defined by the relation $x^3 + y^3 - 2xy = 0$ (the folium of Descartes), find the value of $\dfrac{d^2y}{dx^2}$ at the point $(1, 1)$.

Points of inflection examples:

- $f''(x) = 0$ and there is

1. a point of inflection: $f(x) = x^3 + 3x^2 - x - 3$, $f(x) = \sin(x)$, $f(x) = \arctan(x)$, $f(x) = \dfrac{1}{\ln(x)}$.
2. no point of inflection: $f(x) = a(x-h)^4 + k$.

- ($f''(x)$ is infinite or DNE and there is
  1. a point of inflection: $f(x) = x|x|$, $f(x) = x^{1/3}$, $f(x) = x^{4/5} - x$.
  2. a change in concavity but no point of inflection: $f(x) = \dfrac{1}{x}$).

Examples where the second derivative test:

1. gives a conclusion and is more efficient than the sign test: Any polynomial function.

2. gives a conclusion but is less efficient than the sign test: $f(x) = \dfrac{x-1}{x^2-x+1}$.

3. is inconclusive and the sign test must be used: $f(x) = (x-a)^3 + b$, $f(x) = (x-a)^4 + b$, $f(x) = x^4 \sin^2\left(\dfrac{1}{x}\right)$.

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